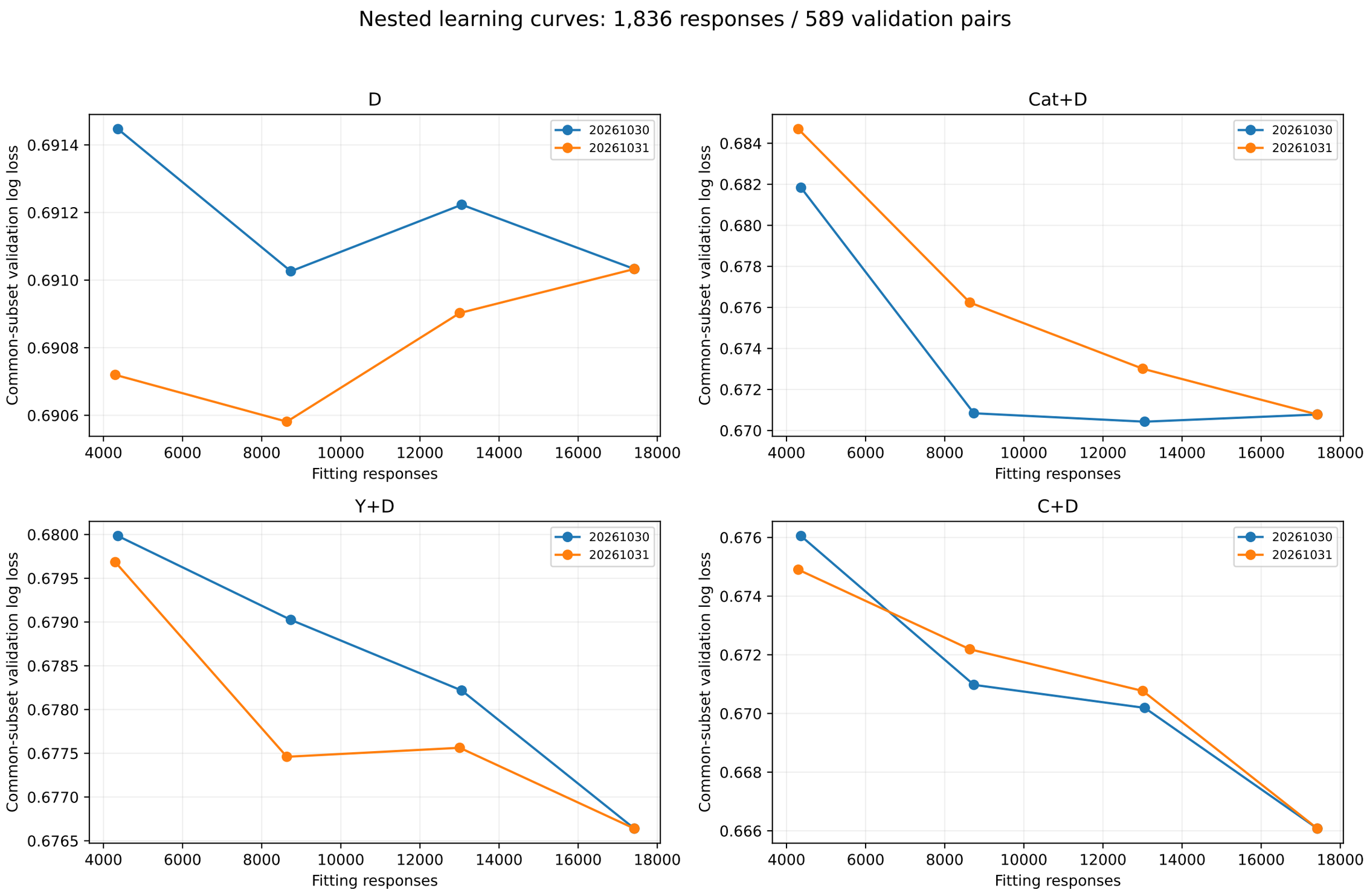
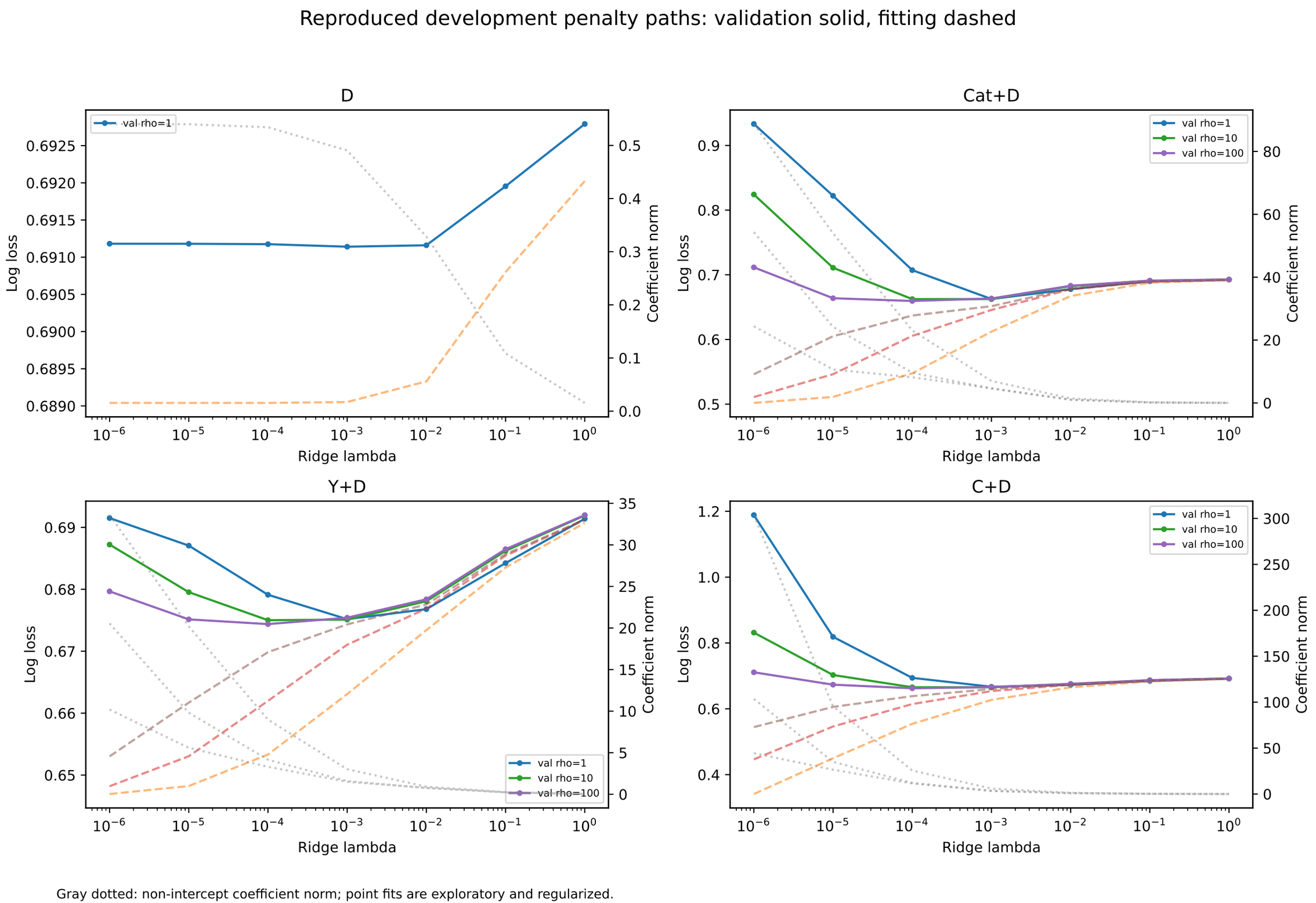
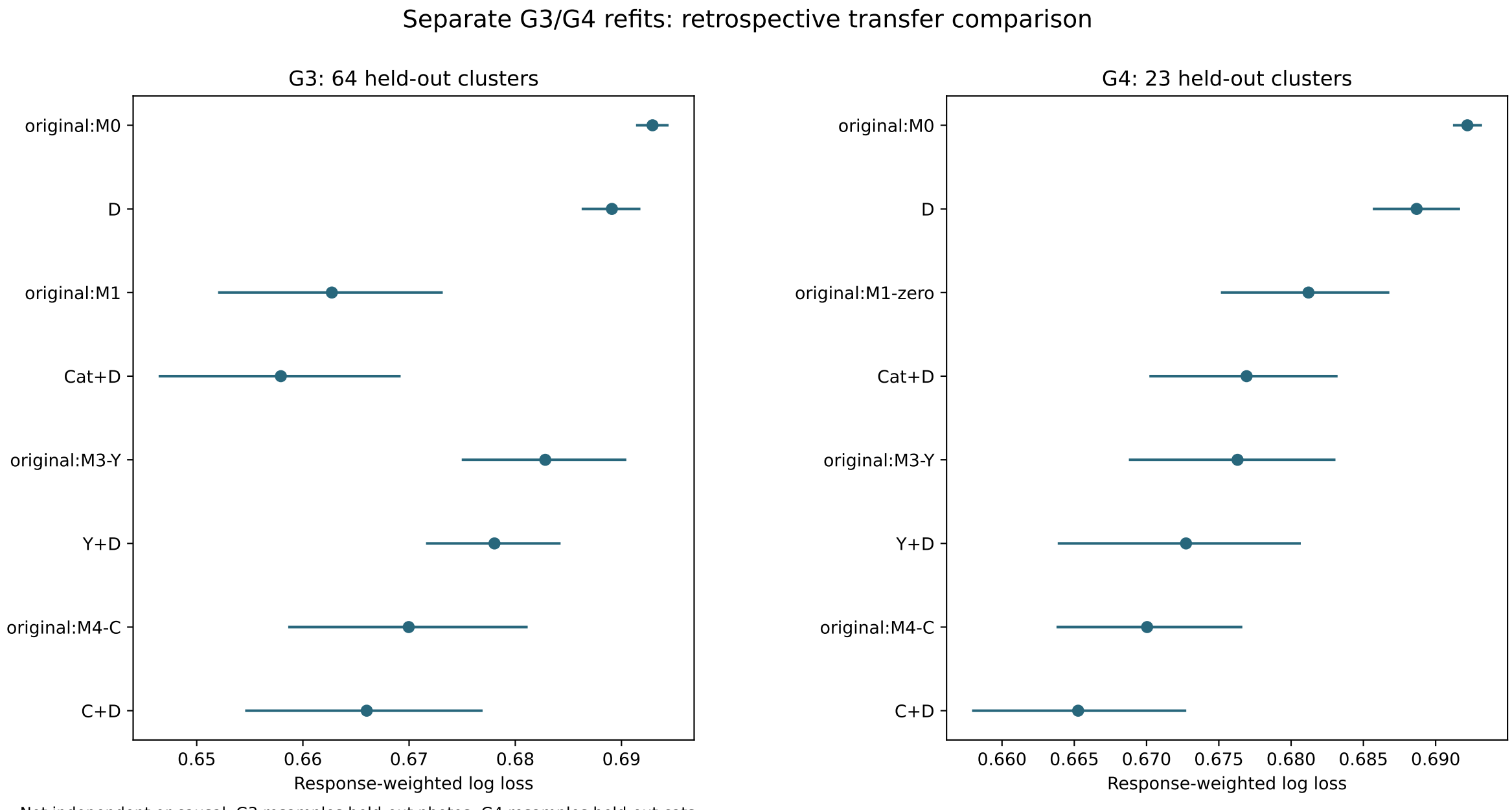


Exploratory; preprocessing, demographics and tuning refit at each size; model fallback can change.


Reproduced development penalty paths: validation solid, fitting dashed
D
Cat+D
Y+D
C+D
val rho=1
val rho=10
val rho=100
Log loss
Coefficient norm
Ridge lambda
Gray dotted: non-intercept coefficient norm; point fits are exploratory and regularized.


Separate G3/G4 refits: retrospective transfer comparison
G3: 64 held-out clusters
original:M0
D
original:M1
Cat+D
original:M3-Y
Y+D
original:M4-C
C+D
0.65
0.66
0.67
0.68
0.69
Response-weighted log loss
G4: 23 held-out clusters
original:M0
D
original:M1-zero
Cat+D
original:M3-Y
Y+D
original:M4-C
C+D
0.660
0.665
0.670
0.675
0.680
0.685
0.690
Response-weighted log loss
Not independent or causal. G3 resamples held-out photos; G4 resamples held-out cats.
G4 Cat+D sets the unseen cat's pooled and demographic-specific score components to zero.